MLS Salary, Roster Construction, and Team Performance

Your Name Teammate 1 Teammate 2

2026-04-12

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```r
library(broom)
library(mgcv)
```

```
## Loading required package: nlme
##
## Attaching package: 'nlme'
##
## The following object is masked from 'package:dplyr':
##
##     collapse
##
## This is mgcv 1.9-4. For overview type '?mgcv'.
```

```r
library(lme4)
```

```
## Warning: package 'lme4' was built under R version 4.4.3
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
##
```

```
## Attaching package: 'lme4'
##
## The following object is masked from 'package:nlme':
##
##     lmList
```

```r
# TODO: Replace these with your actual file paths / data pulls
# salary_data <- read_csv("data/mls_salaries.csv")
# team_data   <- read_csv("data/mls_team_performance.csv")
# player_data <- read_csv("data/mls_player_performance.csv")
```

1 Introduction

Major League Soccer operates under a complex roster-building system, yet teams vary widely in on-field performance. This project studies whether MLS teams perform better by spending more overall, by concentrating salary among a few stars, or by distributing spending more evenly across the roster and across positions.

1.1 Our Central Question

How does salary allocation and roster construction relate to team performance in MLS?

More specifically, we study whether team payroll, salary concentration, and positional spending patterns are associated with outcomes such as expected goal differential, goals added, or points per game.

We use publicly available MLS salary and performance data to evaluate multiple statistical models. Each individual model addresses the same project question from a different angle, allowing us to compare which modeling framework is most informative for understanding roster construction strategy.

1.2 Data

We combine MLS salary information with team- and player-level performance data. • Salary data: TODO • Team performance data: TODO • Player performance data: TODO • Seasons included: TODO

1.3 Unit of analysis

We use two related units of analysis depending on the model: • Team-season for team-level models • Player-season for player-level models

1.4 Variables

1.4.0.1 Outcome variables Potential response variables include: • Team expected goal differential (xGD) • Team goals added differential (g+ diff) • Points per game • Player goals added (g+) • Minutes-adjusted player contribution

1.4.0.2 Main predictors Potential explanatory variables include: • Total payroll • Log payroll • Share of payroll spent on top 3 players • Salary concentration index (Gini / HHI) • Positional spending shares • Average age / experience (if available) • Designated player indicators (if available)

2 Data Cleaning and Preprocessing

```
# TODO: Build your merged datasets here
# Example placeholders:

#team_season <- salary_data |>
#  group_by(team, season) |>
#   summarize(
#     total_payroll = sum(base_salary, na.rm = TRUE),
#     top3_salary_share = sum(sort(base_salary, decreasing = TRUE)[1:3], na.rm = TRUE) / total_
#     .groups = "drop"
#   ) |>
#   left_join(team_data, by = c("team", "season"))

# player_season <- salary_data |>
#   left_join(player_data, by = c("player_id", "season"))
```

Describe any major cleaning steps here

3 Exploratory Data Analysis

Salary distribution

```
# TODO: salary distribution plot
# ggplot(player_season, aes(x = base_salary)) +
#   geom_histogram(bins = 40) +
#   scale_x_continuous(labels = dollar)
```

EDA takeaway: TODO

Team payroll vs team performance

```
# TODO: scatterplot of payroll vs xGD / g+ diff / points
```

EDA takeaway: TODO

Salary concentration vs team performance

```
# TODO: scatterplot of top3 share or HHI vs performance
```

EDA takeaway: TODO

Positional allocation patterns

```
# TODO: stacked bar chart or scatterplot by team/season
```

EDA takeaway: TODO

4 Methods

Shared modeling goal

Across all individual analyses, the goal is to evaluate how roster construction and salary allocation relate to team success. Each group member fits a distinct model, motivated by a different statistical perspective.

5 Individual Analysis 1: Team-level linear regression

Question

Does team performance increase with total payroll, and does salary concentration help or hurt after controlling for total spending?

Model specification

6 fit_lm <- lm(

7 xgd_per_90 ~ log(total_payroll) + top3_salary_share + attacker_share + midfielder_share + defender_share,

8 data = team_season

9)

Assumptions • Linearity • Independence across team-seasons • Constant variance • Limited omitted variable bias

Why this model is appropriate

This provides a simple, interpretable baseline model linking roster spending strategy to performance.

Model evaluation plan • RMSE / MAE • Residual diagnostics • Compare alternative predictor sets

Uncertainty plan • 95% confidence intervals for coefficients • Optional bootstrap confidence intervals

10 Individual Analysis 2: GAM / nonlinear team model

Question

Are the effects of payroll and salary concentration nonlinear?

Model specification

11 fit_gam <- gam(

12 xgd_per_90 ~ s(log(total_payroll)) + s(top3_salary_share) + s(attacker_share) + s(defender_share),

13 data = team_season,

14 method = "REML"

15)

Why this model is appropriate

A GAM allows the relationship between spending and performance to be flexible rather than forcing a strictly linear effect.

Model evaluation plan • Compare against linear regression • Check effective degrees of freedom • Out-of-sample prediction error

Uncertainty plan • Confidence bands for smooth terms • Bootstrap uncertainty if needed

16 Individual Analysis 3: Player-level value model

Question

What kinds of salary allocations generate the most player value per dollar, and how does that aggregate to team-level efficiency?

Model specification

```
# fit_player_gam <- gam(
#   g_plus_per_90 ~ s(log(base_salary)) + position + s(log(base_salary), by = position),
#   data = player_season,
#   method = "REML"
# )
```

Aggregation step

```
# player_season <- player_season |>
#   mutate(predicted_value = predict(fit_player_gam, newdata = player_season))
#
# team_efficiency <- player_season |>
#   group_by(team, season) |>
#   summarize(
#     predicted_team_value = sum(predicted_value, na.rm = TRUE),
#     value_per_dollar = predicted_team_value / sum(base_salary, na.rm = TRUE),
#     .groups = "drop"
#   )
```

Why this model is appropriate

This approach directly addresses the roster construction question by first estimating player-level returns to salary, then rolling them up to the team level.

Model evaluation plan • Player-level predictive performance • Correlation between aggregated team value and observed team outcomes • Compare linear vs nonlinear salary effects

Uncertainty plan • Bootstrap the player-to-team pipeline • Confidence bands by position

17 Results

Individual Analysis 1 Results

```
# TODO: tidy coefficient table / model summary
# broom::tidy(fit_lm, conf.int = TRUE)

# TODO: coefficient plot or fitted vs observed plot
```

Key interpretation: TODO

Individual Analysis 2 Results

```
# TODO: summary(fit_gam)

# TODO: plot smooth terms
```

Key interpretation: TODO

Individual Analysis 3 Results

18 TODO: salary-performance curve by position

19 TODO: team value-per-dollar rankings with uncertainty bars

Key interpretation: TODO

20 Discussion

Our project examined whether MLS teams benefit more from spending heavily overall, concentrating spending in a few stars, or distributing spending more efficiently across the roster.

21 References